Taneem Ullah Jan

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Experience

AI Lead **Oct. 2025 – Present**

ErisAI

- Architect and develop **Synapse**, a research-driven AI infrastructure layer for multi-agent workflows, integrating hybrid retrieval, multi-turn memory, MCP-based tool orchestration, intelligent model routing, Ragas-based evaluation, and Langfuse observability, catching traceability and knowledge-access failures. (Synapse)
- Deployed **Synapse** across multiple enterprise AI applications including legal and investigative platforms, with a LoRA fine-tuned model via Unsloth, processing 30M+ private documents with grounded retrieval and traceable responses.
- Instrumented a durable query ledger across production traffic, turning live queries into re-judgeable evaluation episodes so offline quality measurement stays aligned with real user experience.
- Built **Ownify AI**, a multi-tenant white-label platform on top of Synapse with structural tenant isolation, per-tenant billing and usage metering, and workspace branding for organizations running their own AI product, reaching 7,500+ users and $8.5K+ in early revenue. (Product)

AI Researcher **Sep. 2024 – Oct. 2025**

VOLV AI

- Researched and developed a 3D virtual try-on SDK using neural rendering, GANs, and diffusion models, adopted by fashion brands, driving a 40% increase in user engagement. (OmniFit-3D)
- Developed controllable 2D and 3D digital human systems with pose estimation, generative modeling, and real-time SMPL body-model control at 60 FPS, powering customer-facing virtual try-on and makeup experiences that reduced product return rates by 30%. (FlexiSMPL)
- Prototyped multimodal LLM systems integrating language, vision, and audio for cross-modal reasoning and instruction-following.

Research AI Engineer **Jan. 2023 – Dec. 2023**

BHuman AI

- Built scalable generative video pipelines using neural reenactment, facial motion transfer, and voice cloning, reducing production costs by 40% for a platform serving 90K+ users and enterprise media clients. (Use cases)
- Researched and deployed audio-driven lip-sync models using GANs and super-resolution, improving visual fidelity and establishing it as a core product capability and primary revenue stream.
- Integrated LLMs with persona-driven avatars and developed retrieval-based conversational systems for enterprise clients, combining private knowledge, persona conditioning, and response evaluation for real-world deployments.

Machine Learning Engineer Intern **Aug. 2021 – Nov. 2021**

NAECO Blue GmbH (Profile)

- Developed machine learning models to improve the spatial and temporal resolution of weather data for solar and wind energy forecasting.
- Built a data pipeline that reduced R&D time by nearly 50% and supported forecasting workflows later adopted by 5+ European meteorological agencies.

Projects

- **Synapse**. Research-driven AI system integrating retrieval, memory, routing, evaluation, and runtime controls for reliable model behavior.
- **Ownify AI**. Multi-tenant white-label AI platform built on Synapse, with structural tenant isolation, per-tenant billing, and operational tooling for organizations running their own AI product.
- **DarwinPatch**. Verifier-grounded coding-agent repair framework combining bounded evidence, route-aware search, and regression gates, lifting solve rate from 0.6 to 0.9.
- **PatchProof**. Framework for evidence-grounded patch verification using execution traces, behavioral validation, and correctness analysis beyond unit tests.
- **GateGRPO**. Verifier-grounded reinforcement learning (GRPO) for code repair, using the hard-gate verifier itself as the reward signal in place of a learned reward model.
- **GateDPO**. Verifier-ordered DPO for code repair without a learned reward model, showing task-corpus diversity, not the preference algorithm, controls out-of-distribution generalization.

- **GateRSI**. Recursive self-improvement (RSI) for knowledge agents, routing verified memory, verifier-gated synthetic task generation, and GateDPO weight updates through one verifier, lifting solve rate up to 3.3x with held-out and external-benchmark transfer.
- **GateAdapt**. Measurement-gated admission for self-modifying agent harnesses, using paired measurement, scoped verdicts, and abstention when effects sit inside the noise band.
- **WarmLead**. Cache-liveness scheduler for lead-plus-sidekick LLM agent harnesses, with a cost simulator and live API verification to reduce idle-period token spend.
- **FusionLine**. TypeScript benchmark harness for MCP tool-selection routing, identifying a fusion-boundary rule for reranker combination and cost-verifying results against live provider APIs.
- **DeciForm**. Structured extraction library compiling a Pydantic schema into batched decision-model questions, with calibrated per-field confidence and a generative fallback for only the low-confidence fields.
- **Callona**. Self-hosted speech-to-speech AI receptionist answering business calls in under two seconds, using forced-choice tool routing and deterministic coordinators to keep an 8B model reliable.

Technical Skills

Programming: Python, TypeScript, C++, Bash, MATLAB, SQL/NoSQL (PostgreSQL, MySQL, Redis)

LLM Systems: RAG, Multi-Agent Systems, Autonomous Agent Orchestration, MCP (Model Context Protocol), Function Calling & Tool Use, Multi-Turn Conversation Memory, Prompt Engineering (few-shot, structured output, system-prompt design), Prompt Caching & Context Window Optimization, Human-in-the-Loop Workflows, LLM Guardrails & Safe Action Design, Deterministic Verifier & Grader Design, Inference Optimization (continuous batching, KV cache, latency and cost per token), vLLM, llama.cpp, Ollama

Retrieval & Grounding: Hybrid Retrieval, Vector Search (Qdrant, FAISS, pgvector), Cross-Encoder Reranking, Hallucination Detection (NLI Verification), Structure-Aware Document Chunking, Document Intelligence (OCR, PDF and scanned-document extraction, Docling, RapidOCR), Evidence Provenance & Traceability, Zero-Shot Information Extraction, Ragas, Langfuse

Voice & Real-Time Systems: LiveKit Agents SDK, WebRTC, SIP Trunking, Speech-to-Speech Models, Silero VAD, Neural Vocoders, asyncio (Real-Time I/O)

Frameworks: LangChain, LangGraph, Haystack, FastAPI, Pydantic, Unsloth, TRL, OpenAI API, Anthropic API, Gemini API

ML Fundamentals: PyTorch, Transformers, TensorFlow, Scikit-learn, NumPy, Pandas, Deep Learning, Large-Scale Data Processing, Multimodal Learning, LoRA/PEFT fine-tuning, Quantization, Post-Training (SFT, DPO, Preference Optimization), Reinforcement Learning (PPO, GRPO), Verifier-Based Rewards & Reward Modeling, Synthetic Data Generation, LLM & Agentic Evaluation, Benchmark Task Design (MLE-bench-style), Experiment Design, Offline vs. Online Evaluation Alignment

Infrastructure: Docker, Kubernetes, Git, Linux, CI/CD (GitHub Actions), MLOps (model versioning, monitoring, logging, rollback), Systemd, Weights & Biases, Prometheus, Grafana, Async Processing & Caching (Redis), AWS, Azure, GCP

Computer Vision / 3D: OpenCV, PyTorch3D, MediaPipe, ONNX

Publications

- **Taneem, U. J.**, & Ayesha, N. (2024). Beyond CNNs: Encoded Context for Image Inpainting with LSTMs and Pixel CNNs. International Journal of Innovations in Science & Technology (IJIST) 6(5), 165-179, Special Issue ICTIS 2024.
- **Taneem, U. J.**, & Zakira, I. (2022). HTML Code Generation from Images with Deep Neural Networks. Journal of Engineering and Applied Sciences (JEAS), UET Peshawar.

Education

University of Engineering and Technology Peshawar, Pakistan **Sep. 2018 – Sep. 2022**
Bachelor of Science in Computer Science, CGPA: 3.58/4.0 *Young Undergraduate Researcher Award*